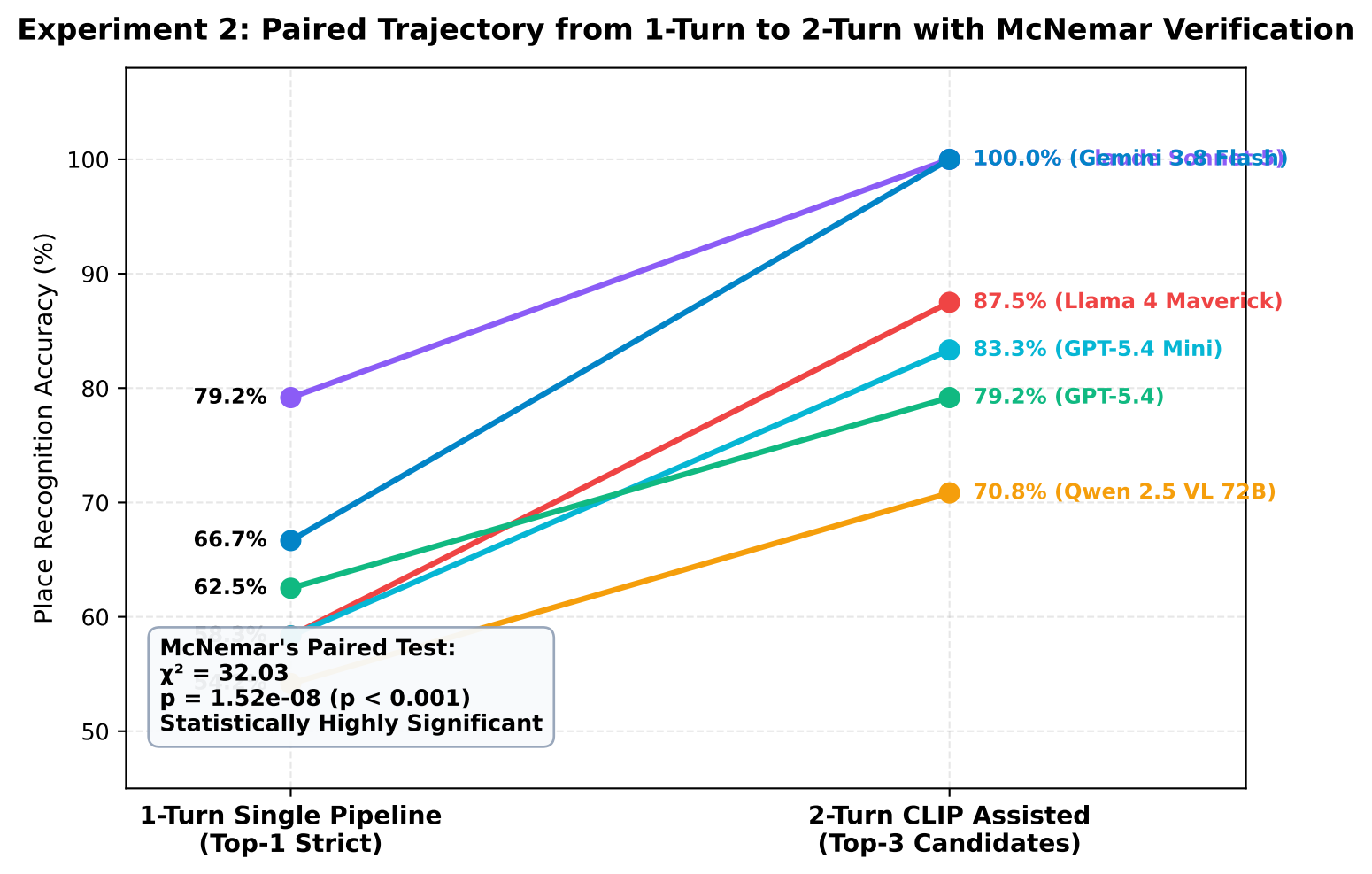
Experiment 2: Paired Trajectory from 1-Turn to 2-Turn with McNemar Verification
Place Recognition Accuracy (%)
100
90
80
70
60
50
79.2%
66.7%
62.5%
100.0% (Gemini 3.1 Flash)
87.5% (Llama 4 Maverick)
83.3% (GPT-5.4 Mini)
79.2% (GPT-5.4)
70.8% (Qwen 2.5 VL 72B)
McNemar's Paired Test:
χ² = 32.03
p = 1.52e-08 (p < 0.001)
Statistically Highly Significant
1-Turn Single Pipeline
(Top-1 Strict)
2-Turn CLIP Assisted
(Top-3 Candidates)